TECHNICAL PORTFOLIO · SUMMER 2027 SOFTWARE ENGINEERING

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Two deep dives

This packet goes deep on two systems I built rather than broad on many. Both are about the same underlying problem: deciding when real-world sensor data can be trusted, when it is late, missing, or possibly fake. That problem shows up in robot autonomy, and it shows up in curating the data that trains AI models.

Project	The hard part	Page
Fusion in Motion Rust GPS/IMU EKF + camera/LiDAR tracking (GT Cloud Robotics VIP)	A drift-invariant consistency check for GPS returning after a 40 s outage; a bounded queue with class-share guarantees; tracing a 3.5x tracking error to delivery-time stamping; avoiding innovation-gate lockout.	2
TrueTake Robot-training video review (Python, OpenCV, CLIP, Claude)	Recovering the clock offset between a camera's video and its gyroscope with a joint lag and axis-mapping search, using it to check footage is genuine and to localize splices. It settled about 69% of clips without human review.	after Fusion in Motion

All numbers come from my own runs and are labeled with their scope (seeds, scenario). Where a result is single-seed or still needs a controlled experiment, the text says so.

Also built

- **Moog Space and Defense Group, Software Engineering Intern (Jun–Aug 2026):** rebuilt maintainable C source for an undocumented legacy binary through black-box testing and fixed its partial-read TCP message framing; built an event-driven WebSocket launcher service enforcing one active operator session; automated RHEL provisioning and hardening with Ansible, from up to 4 hours to about 10–15 minutes (roughly 95% less time per machine) at a 98% STIG pass rate, adopted by 2 more offices.
- **MALD & SWAD:** a Jetson Nano assistive-navigation device fusing YOLO detections with LiDAR depth in a 3-thread condition-variable pipeline, speaking warnings through an iOS app in about 60–130 ms end to end (200 ms target); field-tested with 6 users, 8 devices sold.
- **Metron Finance:** team fintech platform where I worked across the stack, mostly backend, and built the FastAPI ML service (9 REST endpoints, XGBoost scorer, PyTorch LSTM forecaster, online retraining, fallback when a model fails).

GT CLOUD ROBOTICS VIP · RUST · SEP 2026 – PRESENT

Fusion in Motion

Making a robot's GPS/IMU state estimator and object tracker degrade gracefully when the sensor stream stops being clean: overload, late data, and a 40-second GPS outage.

9% / 8%	3.5x	75.9%	~3,900
lower localization / tracking error under overload (3 seeds)	tracking-error bug traced to delivery-time stamping	lower return-window error after a 40 s outage	lines added on my branch, with tests

What I built, in one paragraph

Fusion in Motion is the GT Cloud Robotics VIP's Rust localization and tracking engine (PI: Ethan Boos): a GPS/IMU extended Kalman filter with IMU-bias estimation, plus camera/LiDAR multi-object tracking, driven by a seeded simulator and checked against regression baselines. My branch, `ishaan-sensor-pipeline`, is three commits and about 3,900 added lines across the estimator, a new pipeline module, tests, experiments, and docs. It answers one question: **what should a robot's state estimator do when its sensor stream stops being clean?** That covers three failure modes: more data than the worker can process, data that arrives late, and a sensor that disappears for 40 seconds and then comes back.

- **Sensors (simulated):** IMU at 100 Hz, GPS at 2 Hz (0.25 m σ), camera at 10 Hz (bearing only), LiDAR at 5 Hz (range + bearing).
- **Two clocks on every packet:** `measurement_time_ns` (when the sensor captured it) and `arrival_time_ns` (when the software got it). Most of the bugs below live in the gap between those two numbers.
- **What I added:** a bounded sensor queue with 5 drop policies and per-sensor expiry, a GPS-health monitor, GPS reacquisition logic (variance ramp plus a drift-invariant consistency check), freshness diagnostics, end-to-end and unit tests, regression fixtures, a run-analysis script, and the experiment write-ups.

1. Overload: a bounded queue that degrades on purpose

A real robot's compute can't always keep up with its sensors. I wrote `pipeline::replay`, a deterministic single-worker load model: every packet arrives at its timestamp, waits in a bounded FIFO, and costs a configurable service time per sensor type. When the queue is full, a policy decides who loses. Both the estimator and the tracker then consume only what was actually delivered, with delivery delays applied, so a policy's cost shows up in localization and tracking error rather than just in a drop counter.

The five policies are `drop_newest`, `drop_oldest`, `keep_latest`, `prefer_navigation` (evict camera/LiDAR before IMU/GPS), and `balanced`. Balanced is the interesting one. It guarantees each class (navigation vs. perception) a minimum share of the queue:

```
min_class = min( max( ceil(C · p / 100), C ≥ 4 ? 2 : 1 ), floor(C / 2) )      // p = 25 by
default
max_class = C − min_class
```

On overflow, if the incoming packet's class is under its cap, the victim comes from the other class: specifically the most over-represented *sensor* in that class, and never one that would leave a sensor type with zero queued packets (when C ≥ 4). If the other class is already at its floor, the incoming packet is dropped instead. If the incoming class is at its cap, it evicts the oldest queued packet from its own sensor (or its own class). FIFO order is preserved for everything that stays. Each sensor also gets an optional maximum queue wait, so a stale packet is expired rather than served late.

3-seed sweep, 15 s overload case	balanced	drop_newest
Ego localization RMSE (capacity 8)	**0.185 m**	0.204 m
Track RMSE, estimated ego (capacity 8)	**0.240 m**	0.262 m
Track RMSE, truth ego (capacity 8)	**0.026 m**	0.031 m
Track RMSE, truth ego (capacity 32)	**0.026 m**	0.033 m (prefer_navigation: 0.087 m)
What balanced gave up	197 IMU packets per run at capacity 8; zero GPS, camera, or LiDAR	

That's a 9% localization and 8% tracking improvement from shedding the one stream that is most redundant at 100 Hz. The lesson I wrote into the docs: `prefer_navigation` sounds safe, but starving perception made tracking more than 3x worse than balanced at capacity 32.

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2. The 3.5x bug: late data stamped as new data

When I ran the queue against the GPS-recovery scenario, localization barely moved, but object-tracking error jumped from **0.035 m to 0.123 m (3.5x)**. The queue had dropped 748 of 6,000 IMU packets (12.5%) and no detections, so data loss wasn't the cause.

The cause was time. A camera or LiDAR detection that waited in the queue was being handed to the tracker stamped with its *delivery* time. The tracker then predicted each object forward to the wrong instant before associating and updating, so every update applied an old observation as if it were current. For a target moving at speed v with queue delay d, that injects a position bias on the order of $v \cdot d$ into every update. That is the classic out-of-sequence-measurement problem, and it is invisible in drop statistics.

The engine already had a tracker timing-compensation option that keeps each detection's capture time, but it was off by default, so nothing had ever exercised it under queue delay. Turning it on returned track error to **0.035 m** with no false-track samples. I made it the default for the overload scenario and documented why the GPS-recovery experiment itself stays queue-free when isolating the filter. The

general rule I took from it: **a queue changes when data arrives, never when it happened**, and every consumer must be explicit about which of the two clocks it is using.

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3. A 40-second GPS outage, and the gate that locks GPS out

The recovery challenge drives the vehicle through a left and then right turn with GPS absent from 10 s to 50 s, leaving 10 s to recover. During the outage the filter dead-reckons on biased IMU data (simulated accelerometer bias 0.025 m/s², gyro bias 0.003 rad/s), so position error grows to meters (outage-window RMSE 3.06 m, about 21.5x the continuous-GPS control).

The trap. The EKF rejects any GPS fix whose normalized innovation exceeds a 3-σ gate. If the filter's covariance grows more slowly than its true error during the outage, the very first fixes after the outage look like outliers and get rejected. Each rejection keeps the estimate un-corrected, so the next fix is rejected too. I wrote a unit test (`gps_return_can_be_gated_out_after_loss`) that reproduces exactly this: an accelerometer bias the basic filter doesn't model, 20 s without GPS, and the returning fix is rejected by the 3-σ gate.

The variance ramp. After a stale gap (no fix for 1.5 s), returning fixes are given inflated measurement noise $R' = s \cdot R$, where s decays quadratically from an initial multiplier to 1 over a ramp window:

```
s(t) = 1 + (s0 − 1) · (1 − p)²,     p = clamp((t − t_return) / T, 0, 1)
```

One correctness detail matters here. The inflated R' has to be used consistently in the innovation covariance, the Cholesky solve for the Kalman gain, and the Joseph-form covariance update ($P^{+} = (I - KH)\,P\,(I - KH)^{T} + K\,R'\,K^{T}$), while the outlier gate still uses the receiver's nominal R. If you inflate R for the gain but leave the nominal value in the Joseph term, the filter reports less uncertainty than it actually has. I changed all three places together.

Tuning against a stated priority. The original ramp (1000x over 3 s) trades accuracy for smoothness: it shrinks the biggest correction jump but keeps the estimate wrong for longer. On the same seed, queue settings, and input stream:

Recovery setting	Return RMSE (50–53 s)	Post-recovery RMSE	Largest return step	Final error
1000x over 1 s (prior)	1.078 m	0.108 m	3.633 m	0.0300 m
800x over 0.5 s	0.885 m	0.106 m	3.912 m	0.0303 m
1.5x over 0.5 s (chosen)	**0.260 m**	0.104 m	5.776 m	0.0311 m
Abrupt return (control)	0.252 m	0.103 m	5.777 m	0.0315 m

The chosen setting cut return-window error **75.9%** versus the prior ramp and lands within 0.008 m of the abrupt control. I documented the cost too: it gives back the smaller correction jump. That's a deliberate trade for accuracy, and it's a one-seed result, so I labeled it that way.

4. The hard part: a drift-invariant check for returning GPS

A variance ramp treats every returning fix the same. What you actually want is to trust a good fix immediately and stay skeptical of a bad one (multipath, a jump, a spoofed position). The obvious test, "is the fix close to my estimate?", is useless right after an outage, because the estimate has drifted by meters. A perfectly good fix looks just as far off as a bad one.

My adaptive check compares *motion* instead of *position*. For two successive returning fixes z_{k-1} and z_k, with filter estimates $\hat{x}_{k-1}$ and the IMU-propagated prediction $\hat{x}_k^-$:

$$d = \| (z_k - z_{k-1}) - (\hat{x}_k^- - \hat{x}_{k-1}) \|$$
$$\text{consistent} \iff d \le 3 \cdot \max\left(\sqrt{\sigma^2_{k-1} + \sigma^2_k},\ 0.25\ \text{m} \right), \quad \text{with } 0 < t_k - t_{k-1} \le \text{stale limit}$$

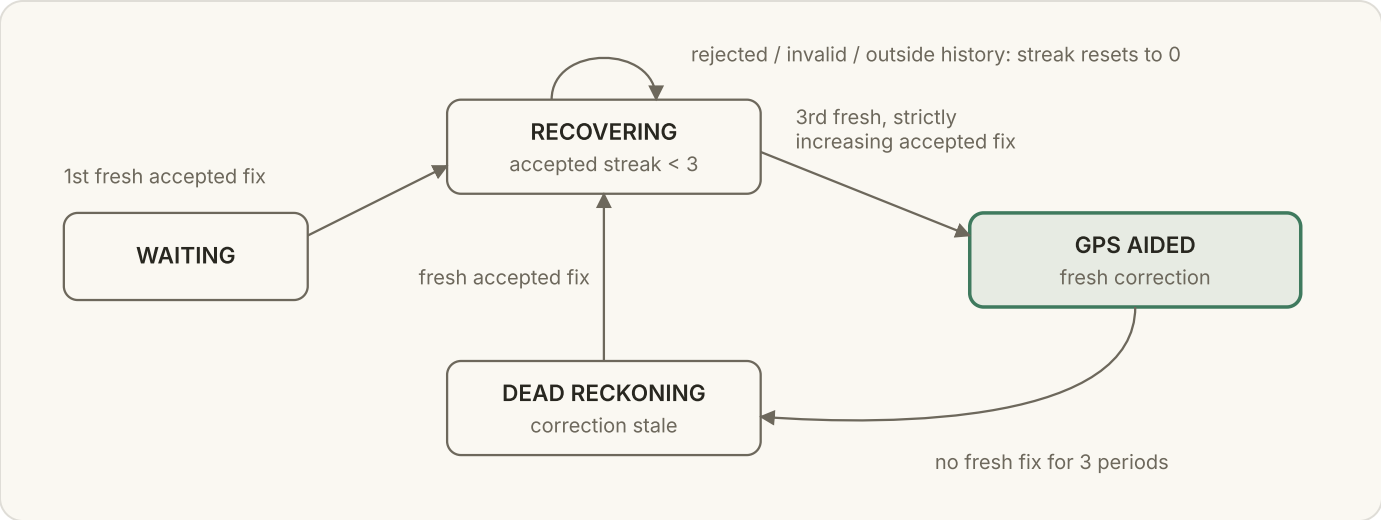
Why this works: dead-reckoning error over a 40-second outage is mostly an accumulated offset that changes slowly. Over the half second between two fixes, that offset is nearly constant, so it cancels when you subtract the two positions. What's left compares how far GPS says the robot moved against how far the IMU says it moved over the same short interval, which the IMU measures well even when its absolute position is meters off. The threshold is 3-σ of the difference of two independent fixes (variances add), with a 0.25 m floor so a too-confident receiver can't make the test impossible to pass.

- A consistent pair returns GPS to nominal weight at once.
- A mismatch holds a 5x variance multiplier and keeps checking.
- The first fix after the gap, which has nothing to pair with yet, gets the short ramp.
- Only applied fixes advance the pair. A rejected fix resets it, so one bad fix can't vouch for the next.

Measured effect (queue-matched, same seed) versus no ramp: overall RMSE −1.8%, outage-window RMSE −1.9%, final error −46%, at a cost of +5.2% return-window RMSE. In the queue-free end-to-end test it cut final error from 0.0418 m to 0.0263 m. The honest caveat, which is also in the repo: the mismatch branch still needs a controlled GPS-jump experiment before I'd call it robust.

5. Knowing when to trust the estimate: the GPS-health monitor

A filter will happily report a position whether or not it has been corrected recently. Operators need to know which situation they are in, so I built a monitor that replays every GPS decision (accepted, rejected, invalid, outside the filter's history buffer) in arrival order and reconstructs a four-state machine. It observes the filter and never changes its tuning or state.



- **Stale** means no fresh correction for three nominal GPS periods (1.5 s at 2 Hz).
- **Re-entering GPS AIDED** requires three fresh accepted fixes with strictly increasing measurement times. A rejected, invalid, or outside-history fix resets the streak to zero, so a burst of bad or replayed packets can't fake recovery.
- **Two different ages.** Receipt age is "how long since any packet arrived"; correction age is "how old is the newest measurement the filter accepted." A delayed or out-of-order packet updates the first but not the second, so late data never looks like fresh data.
- **Output:** every fix decision with its normalized residual, health samples over time, and a notification with recommended operator action only on state changes, saved to `gps-health.json` and shown live in the Rerun dashboard.

The unit test walks a 60-second run with a 40-second outage and a rejected first return fix, and asserts the exact sequence: dead reckoning at 49 s, still dead reckoning when the rejected fix arrives at 50 s (receipt age 0, correction age 41 s), recovering at 51–52 s, aided at 53 s, and exactly five notifications.

How I kept it honest

- **Regression baselines:** every experiment's metrics live in a fixtures file. Tests fail if a change moves them, so improvements are always deliberate and reviewed.
- **Paired comparisons:** policy comparisons use the same seeds, service times, and input streams, so differences come from the policy and not the noise.
- **Window metrics, not just totals:** outage [10, 50) s, return [50, 53) s, and post-recovery [53, 60) s are scored separately, because aggregate RMSE hides exactly the moments that matter.
- **Tooling:** `analyze-runs.py` exports saved-run metrics and deltas against a reference run to CSV, with its own tests.
- **Stated limits:** recovery results are single-seed and simulated. My advisor's next request is to characterize real open-source sensor data, since simulated noise is cleaner than reality.

What I'd do next

- Multi-seed recovery sweeps with confidence intervals, plus outages of different lengths and motion profiles.
- Inject controlled GPS jumps and multipath to test the mismatch branch of the consistency check.
- Replace drop-and-compensate with true out-of-sequence measurement handling in the EKF (retrodiction from a state history buffer).
- Re-run the queue and recovery experiments on real GNSS/IMU logs, such as UrbanNav, to see how much of this survives real noise.

PYTHON · OPENCV · CLIP · CLAUDE API · FASTAPI · RERUN · NOV 2026

TrueTake

A pipeline that reviews first-person robot-training videos against a lab's requirements. Its core is recovering the clock offset between a camera's video and its gyroscope, which also checks that the footage is genuine and unedited.

~69%	0.43–0.67	±2 s	235
of clips settled without human review (18 of 26)	real gyro match vs 0.02–0.14 for swapped data	joint lag and signed axis-mapping search	tests passing

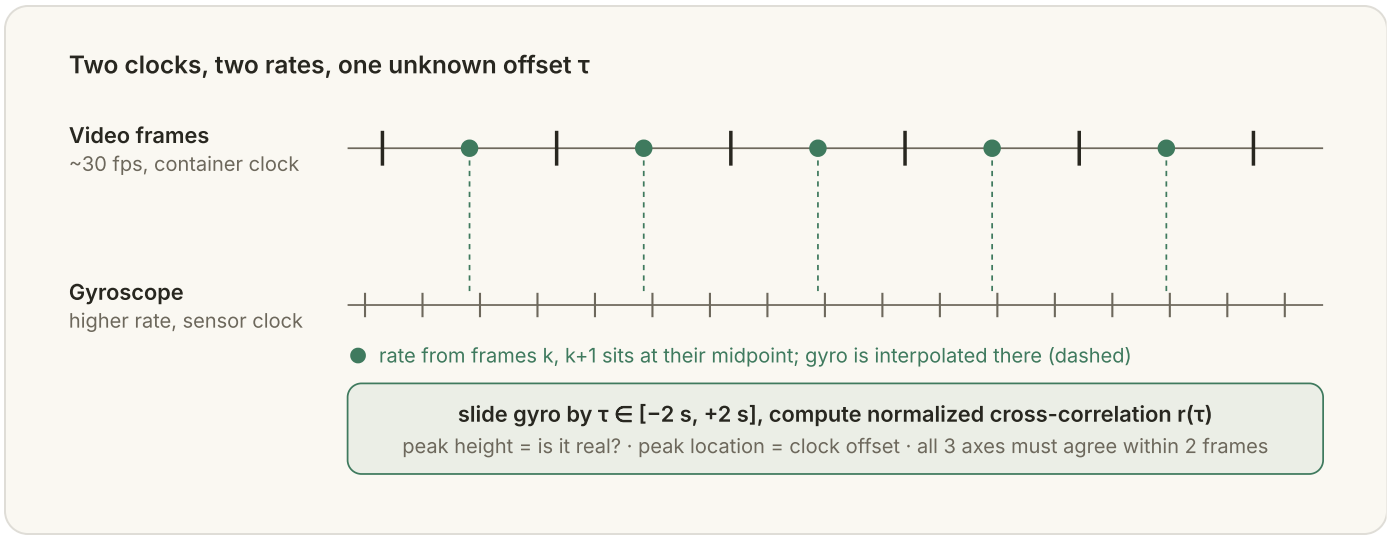
The problem

Robotics labs collect first-person task videos, such as folding a shirt or washing dishes, as robot-training data. Many clips turn out to be unusable: the task isn't actually done, hands leave the frame, the camera is idle, a clip duplicates another, or the footage isn't what it claims to be. TrueTake reviews a batch against a lab's written requirements (a YAML contract) and gives every clip one of four outcomes, **APPROVED, NEEDS REVIEW, REJECTED, or DUPLICATE**, each with timestamped reasons and usable minutes.

- **Quality:** format and privacy rules, hand presence, idle time, and whether the requested goal and process are visible. Measured rules are decided by formulas against the contract's targets and review bands.
- **Verification:** does the camera motion in the video match the camera's own gyroscope? (the main section below)
- **Value:** ranks approved clips by how much new behavior they add, flags duplicates, and suggests what to film next.
- **Keeping the AI honest:** Claude must return evidence before any yes/no/unsure goal answer. A validator checks every cited frame ID, timestamp, and evidence reference; Claude never overrides a formula, and unsure or invalid evidence becomes NEEDS REVIEW rather than a silent pass.

Verification is a clock-synchronization problem

A head-mounted camera records video and gyroscope data into separate streams, on separate clocks, at different rates. If the footage is genuine and unedited, the way the picture rotates must match the way the camera rotated. Testing that first requires a harder answer: **what is the time offset between the two streams?** One computation gives both a match score (do they agree?) and a clock offset (how to align them), which labs need anyway to sync video with sensor data.



How it works, step by step

1. Camera rotation from pixels. For each pair of sampled frames: pick up to 200 corners (`goodFeaturesToTrack`), track them with pyramidal Lucas–Kanade optical flow (21 px window, 3 pyramid levels), and fit a rotation-plus-translation model with RANSAC (2 px reprojection error) so hands and moving objects are rejected as outliers. A pair is discarded if fewer than 20 points track, under half are inliers, or the frames are more than 0.2 s apart. The fit gives pan and tilt in pixels/s and roll in rad/s.

2. Time-stamp each measurement at the right instant. A difference between two frames measures the rate at their *midpoint*, so each flow sample is stored at the pair's actual midpoint time. Stamping it at either frame would build a half-interval error into the signal, which the cross-correlation would then misreport as clock offset.

3. Put both signals on one grid without inventing data. Video rates are resampled onto a 0.1 s grid and smoothed with a centered 0.2 s window. Gyro values are interpolated onto the same grid, shifted by each candidate lag. Interpolation refuses to bridge any gap longer than 0.15 s, including after shifting, so a dropped stretch of sensor data stays missing instead of being filled with a straight line that could fake agreement.

4. Search lag and axis mapping together. For every lag from −2 s to +2 s in 0.1 s steps (41 lags), the code computes the Pearson correlation between each video axis and each gyro axis, wherever at least 50 valid pairs cover at least 5 s and both signals actually move. Which gyro axis is pan, tilt, or roll depends on how the camera is mounted, so it searches every signed permutation of the three axes. The score is the mean signed correlation over at least two informative axes, and the best (lag, mapping) pair wins:

```
r(τ) = Σₜ (vₜ − v̄)(gₜ₊τ − ḡ)  /  ( √Σₜ(vₜ − v̄)² · √Σₜ(gₜ₊τ − ḡ)² )
score(τ, π, s) = mean over informative axes i of  sᵢ · rᵢ,π(ᵢ)(τ)
```

Because Pearson correlation ignores scale, the camera's focal length is never needed: for small rotations, image shift is roughly focal length × angular rate × Δt, so the video signal is a scaled copy of

the gyro signal and the unknown scale drops out.

5. Calibrate once, then lock. An exploratory search can find a mapping that fits a bad clip by chance, so it never earns a pass. A calibrated profile, fitted only on training clips, fixes the signed mapping (for the GoPro setup in our data: pan↔gz, tilt↔gy, roll↔gx, all positive) and the acceptance threshold (0.29). Profiles are validated: a provisional profile can't carry a threshold, and a calibrated one must list the training clips it came from.

6. Make the axes agree. The two strongest axes must each peak at nearly the same lag (within 0.2 s). Only the two strongest vote, because camera stabilization can flatten one axis, and a weak axis shouldn't veto two strong, independent ones.

7. Refuse to guess. No verdict is issued when the gyro barely moved (standard deviation under 0.02 rad/s), when overlap is too short, when the best lag sits at the ±2 s search boundary (the true offset may be outside the window), or when the score falls inside a review band (a close call). Those clips go to NEEDS REVIEW. Offsets over 1 s are flagged.

8. Localize edits. The same measurement runs on 5-second windows starting every second. A window where agreement collapses inside an otherwise matching clip marks where a splice is. Windows are evidence for the reviewer; they don't decide the verdict, because overlapping windows aren't independent tests.

Results

Check	Result
Real gyro vs. video (7 training pairings, one fixed mapping)	Scores **0.43–0.67**
Swapped gyro from another clip (same pairings)	Scores **0.02–0.14**, a clear gap below the 0.29 threshold
Spliced clip	Rejected at **0.153 < 0.29**; window evidence marks the cut at 0:30
Full 26-clip run	5 approved, 8 needs review, 12 rejected, 1 duplicate; 3.3 approved usable minutes, including one sensor-verified clip
Human review avoided	**18 of 26 clips (about 69%)** settled automatically, so reviewers only had to look at 8. Run overnight, that meant people reviewed only the leftovers, outside the lab, with screened data ready each morning.
Test suite	235 tests passing, plus real-clip CLI checks and validated Rerun recordings

The gyro scores are calibration and development results on a small clip set, not held-out accuracy, and the project says so. Gyro agreement is supporting evidence, not proof of authenticity. Uncached processing runs roughly 2.5 minutes per clip on a laptop CPU.

Value: which clips actually teach a robot something new

Only approved clips are ranked, so "unusual" can't just mean "broken." Each pair of clips gets three distances:

- **Scene:** averaged CLIP embeddings; distance is 1 − cosine similarity.
- **Hand motion:** hand paths resampled to 64 steps; distance is dynamic time warping.
- **Strategy:** Claude's labeled step sequence for the task; distance is normalized edit distance.

Each view is divided by its median and averaged. Ranking is greedy farthest-point selection: start from the most representative clip, then repeatedly add the one farthest from everything already chosen. Near-identical clips are caught separately by comparing perceptual hashes of 8 frames, including a mirrored copy, so a flipped re-upload still counts as a duplicate.

What I'd push further

- A held-out test set with complete human labels, so accuracy can be reported instead of development scores.
- Sub-sample clock offset by fitting a parabola through the correlation peak instead of snapping to the 0.1 s grid.
- Rolling-shutter correction, since each image row is captured at a slightly different time during fast motion.
- Automatic escalation to a stronger model for unsure goal answers, which is designed but not yet implemented.